

LAB #8: Setting Up VLAN (part 2)

Lab Requirements:

- Packet Tracer tool
- Lab Document
- Windows/Mac OS Host

1. Overview

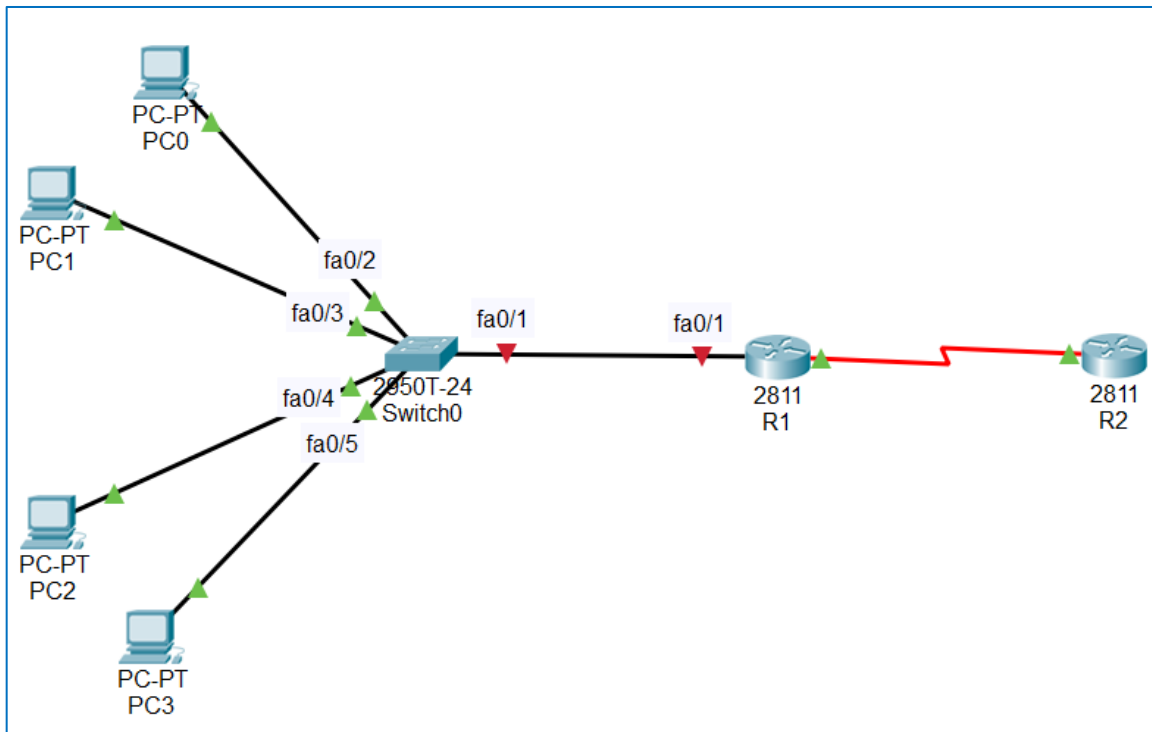
In this in lab study, you will create 2 static VLANs. Your goal is to increase security by creating VLANs. A VLAN is a group of devices on one or more LANs that are configured to communicate as if they were attached to the same wire, when in fact they are located on a number of different LAN segments. Because VLANs are based on logical instead of physical connections, they are extremely flexible. VLANs define broadcast domains in a Layer 2 network. Layer 2 switches create broadcast domains based on the configuration of the switch.



Cisco 2950 Series Switch Rear View

2. Simulation

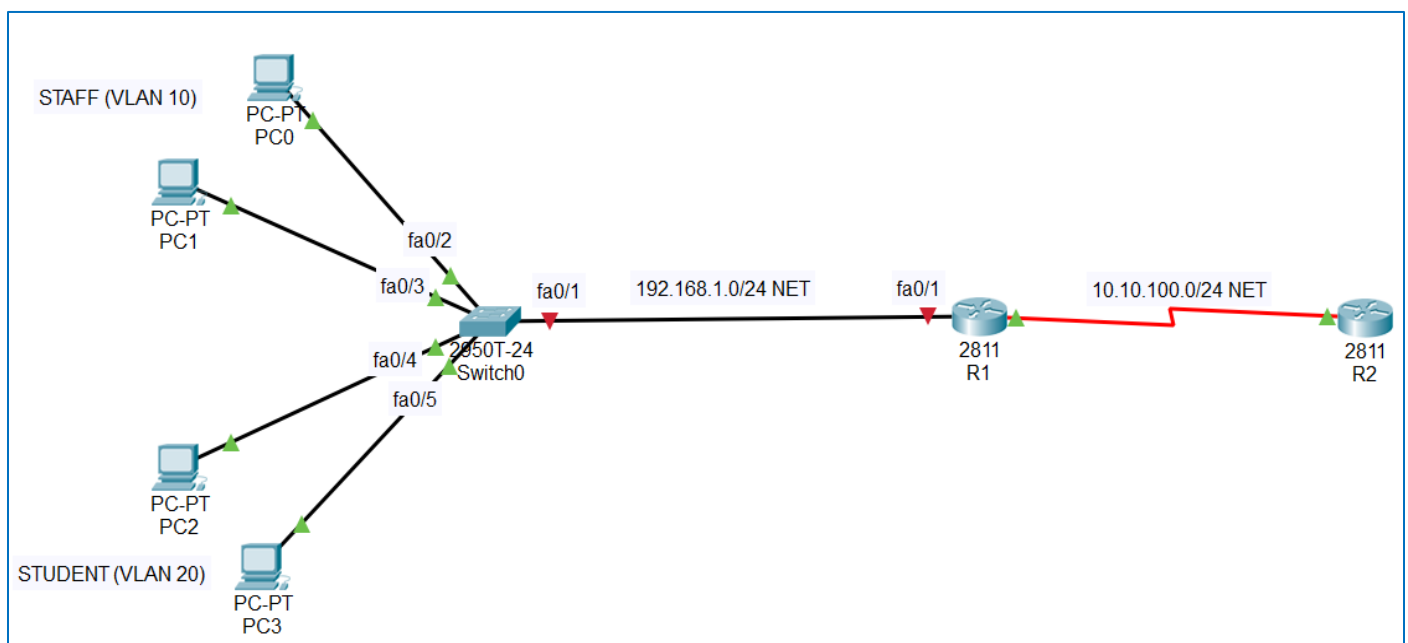
We will continue from our previous lab that we have connected North American University campus routers. Open the saved file. Drag "2950T" switch and connect it to R1 with "Copper Straight-Through" from Switch FastEthernet0/1 to R1 FastEthernet0/1. Then drag 4 PCs and connect them to the switch by order to the switch interface FastEthernet 0/2 – 0/5. Now your network should look like as shown below. Between switch and R1 link is down because R1 fa0/1 interface is not enabled yet.



Copper Straight-Through cable (RJ45/Cat6/Ethernet cable)

Ethernet cables can be wired as **straight through** or **crossover**. The straight through is the most common type and is used to connect computers to hubs or switches. They are most likely what you will find when you go to your local computer store and buy a patch cable. Crossover Ethernet cable is more commonly used to connect a computer to a computer and may be a little harder to find since they aren't used nearly as much as straight through Ethernet cable.

Willowbrook LAN IP is 192.168.0.0/16. We will divide our LAN into two VLANs because we don't want Student department network to interfere with Staff department network. Staff department will be configured VLAN id 10 and Student VLAN id 20. PC 0-1 are Staff department and PC 2-3 are Student department as shown below.



- 1) Assign IP address to R1 FastEthernet0/1 (gateway) and enable the interface with "no shutdown" command.

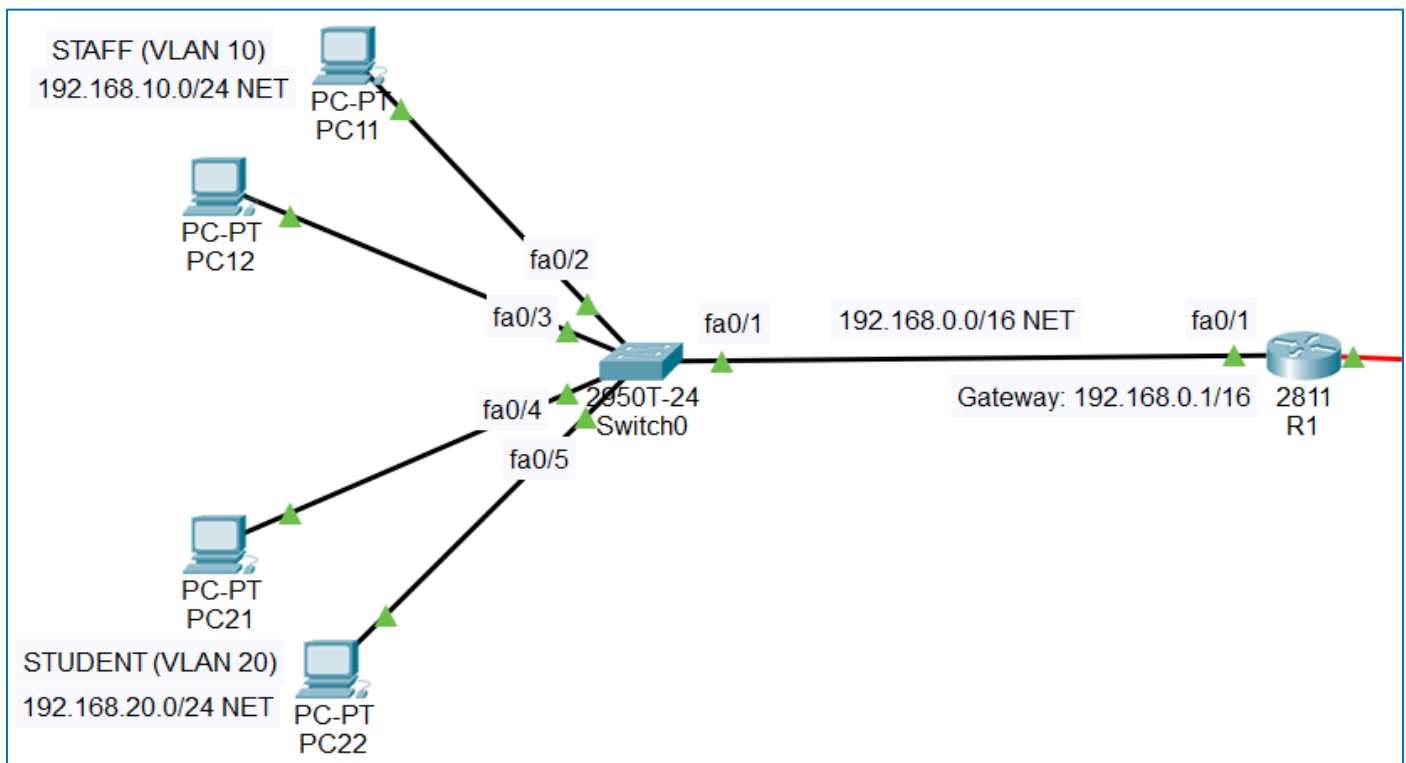
```
Willowbrook>enable
Willowbrook#configure terminal
Willowbrook(config)#interface fa0/1
Willowbrook(config-if)#ip address 192.168.0.1 255.255.0.0
Willowbrook(config-if)#no shutdown
```

```
Willowbrook>enable
Willowbrook#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Willowbrook(config)#interface fa0/1
Willowbrook(config-if)#ip address 192.168.0.1 255.255.0.0
Willowbrook(config-if)#no shutdown

Willowbrook(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Now between R1 and switch link light must be turned in green.

We will subnet 192.168.0.0/16 LAN network such as 192.168.10.0/24 for VLAN 10 and 192.168.20.0/24 for VLAN 20.



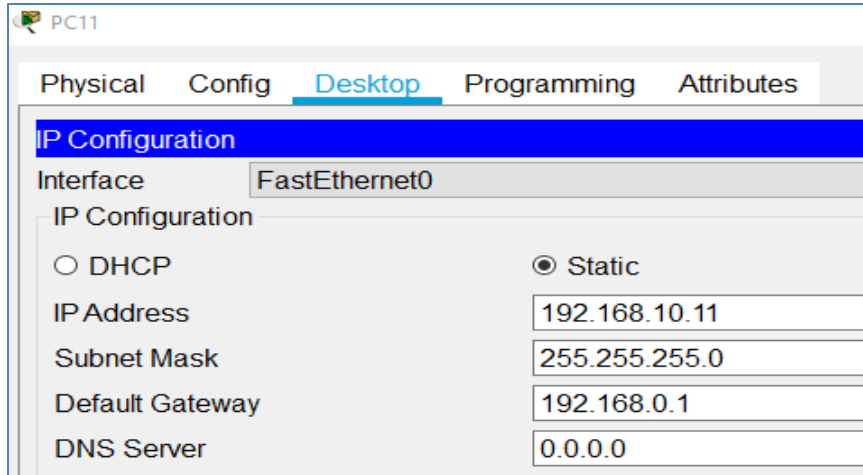
- 2) Rename PC names as shown above. PC 0, 1 to 11, 12 and PC 2, 3 to 21, 22.
- 3) Place text notes as shown above to be clear. You can add notes by pressing “N” key.
- 4) Assign IP addresses to PCs. Click on PC icon and go to “Desktop” tab then select “IP Configuration”.

PC11 to 192.168.10.11/24

PC12 to 192.168.10.12/24

PC21 to 192.168.20.21/24

PC22 to 192.168.20.22/24



The screenshot shows the 'PC11' configuration window with the 'Desktop' tab selected. Under 'IP Configuration', the 'FastEthernet0' interface is configured with the following settings:

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IP Address	192.168.10.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.1
DNS Server	0.0.0.0

Now all PCs should be able to ping each other and gateway and R1's WAN except R2. Test them all.

5) Now it's time to create VLANs. Open switch CLI. Create VLAN 10 and 20 and enter the names. Entering name is an optional but suggested to avoid confusions in the future. After creating VLANs verify entries with "**show vlan brief**" command. By default, all ports are assigned to VLAN1.

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 10
Switch(config-vlan)#name STAFF
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name STUDENT
Switch(config-vlan)#end
Switch#show vlan brief
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name STAFF
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name STUDENT
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	STAFF	active	
20	STUDENT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

6) Assign fa0/2, fa0/3 to VLAN 10 and fa0/4, fa0/5 to STUDENT VLAN. Then verify if ports are assigned to proper VLANs with “show vlan brief” command.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface f0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface f0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface f0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#end
```

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	STAFF	active	Fa0/2, Fa0/3
20	STUDENT	active	Fa0/4, Fa0/5

Now PCs should be able to ping **only** 2nd PC on same VLAN. Make sure that any PC should not be able to ping R1 or other VLAN PCs. Any PC can't ping to gateway because R1 Fa0/1 is assigned to VLAN 1 (different virtual network) by default.

7) Configure switch fa0/1 port to trunk mode

```
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan add 10
Switch(config-if)#switchport trunk allowed vlan add 1
Switch(config-if)#switchport trunk allowed vlan add 20
Switch(config-if)#end
```

8) Configure R1 as shown below to enable trunking between Switch and Router

```
Willowbrook#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Willowbrook(config)#int fa0/1
Willowbrook(config-if)#no ip address
Willowbrook(config-if)#no shut
Willowbrook(config-if)#exit
Willowbrook(config)#int fa0/1.10
Willowbrook(config-subif)#encapsulation dot1q 10
Willowbrook(config-subif)#ip address 192.168.10.1 255.255.255.0
Willowbrook(config-subif)#no shut
Willowbrook(config-subif)#exit
Willowbrook(config)#int fa0/1.20
Willowbrook(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.20, changed state to up

Willowbrook(config-subif)#encapsulation dot1q 20
Willowbrook(config-subif)#ip address 192.168.20.1 255.255.255.0
Willowbrook(config-subif)#no shut
Willowbrook(config-subif)#exit
Willowbrook(config)#end
Willowbrook#
%SYS-5-CONFIG_I: Configured from console by console

Willowbrook#write
Building configuration...
[OK]
Willowbrook#
```

Now change VLAN 10 PCs gateway address to 192.168.10.1 and VLAN 20 PCs to 192.168.20.1.

You should be able to ping all PCs now.

Congratulations! You have completed the Lab part II.

3. References

- [1] http://www.cisco.com/public/scc/compass/3600/information_files/inf_3600_interface_numbering.htm#fig_3
 - [2] http://www.cisco.com/en/US/products/hw/routers/ps274/products_data_sheet09186a0080091b89.html
 - [3] <http://www.cablesandkits.com/cisco-1port-fast-ethernet-network-module-nm1fetx-p-867.html>
 - [4] A Practical Guide to Advanced Networking, 3rd Edition
 - [5] http://www.cisco.com/c/en/us/td/docs/ios/12_2/configfun/command/reference/ffun_r/frf001.html#wp1017386
 - [6] http://www.cisco.com/c/en/us/td/docs/ios/12_2/configfun/configuration/guide/ffun_c/fcf001.html
 - [7] http://www.cisco.com/c/en/us/td/docs/security/asa/asa90/configuration/guide/asa_90_cli_config/basic_hostname_pw.html
 - [8] http://www.cisco.com/c/en/us/td/docs/ios/12_2/ip/configuration/guide/fipr_c/1cfipadr.html
 - [9] http://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960/software/release/12-2_55_se/command/reference/2960_cr/cli3.html
- <http://www.cisco.com/c/en/us/support/docs/ip/routing-information-protocol-rip/13730-ext-ping-trace.html>